

Efficient gate-set extensions for fault-tolerant qudit quantum computation

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Fault-tolerant quantum computation requires a finite universal gate set whose members have cheap physical realizations. We introduce SWIFTBOT, an agentic workflow that numerically ranks finite subgroups of $SU(d)$ extended by a single qudit gate, using the Quantum Circuit Overhead Q_T as the figure of merit. At $d = 3$ we produce the first systematic Q_T ranking across three Σ -series subgroups, finding that Haar-random extensions on the largest approach the Kesten–McKay floor within 32%, while structured diagonal and Howard–Vala extensions sit factors of 3–70 above it. Applied to gate sets relevant for lattice-gauge theory simulation, a non-canonical rational-phase extension delivers $6.8\times$ lower mean synthesis depth than canonical Clifford+ T , demonstrating that the best general-purpose extension and the best target-specific extension can differ by nearly an order of magnitude. The pipeline, registered groups, fault-tolerance catalog, and reproduction data are released with the paper.

I. INTRODUCTION

Fault-tolerant quantum computation requires a finite universal gate set whose members have cheap physical realizations. The tension between these requirements is sharp: the Eastin–Knill theorem [1] forbids any nontrivial stabilizer code from admitting a universal *transversal* gate set. Practical designs therefore split the gate set into a transversal subgroup $\mathcal{C} \subset SU(d)$ (usually a finite subgroup the error-correcting code protects cheaply) and one or more non-transversal extension gates g made fault-tolerant by magic-state distillation [2, 3] or by code switching between codes with complementary transversal structure. Each extension gate g comes with its own cost: a dedicated magic state, a code that supports it transversally, a distillation protocol, or some combination of the three.

Why qudits. The qubit version of this story is well-developed but increasingly constrained: Eastin–Knill rules out any universal transversal stabilizer construction, and the best-known bypass—magic-state distillation of T on Clifford+ T —is the overhead bottleneck of every qubit fault-tolerance resource estimate. Qudit hardware sidesteps some of this by exposing multiple native levels in a single degree of freedom [4, 5]: superconducting cavities with multi-mode Fock spaces [6, 7], trapped-ion d -level systems, and solid-state spin- S qudits all avoid the binary-encoding overhead that qubits impose on inherently higher- d physics. Lattice-gauge simulation makes the case sharpest: the gauge fields of $SU(N)$ lattice theories live naturally on d^{N^2-1} -dimensional registers [10? ? ? ? –13], so a qudit- or quocit-based architecture is a better fit than packing them into qubits; this programme has progressed from formal proposals to experimental trapped-ion qudit realisations of $U(1)$ and $SU(2)$ gauge dynamics [? ?]. A complementary strand of work adapts

the error-correction side to the gauge-redundant Hilbert space. Halimeh *et al.* [?] protect gauge invariance via single-body energy penalties; Rajput, Roggero, and Wiebe [?] correct one-qubit errors on $\mathbb{Z}_2$ and truncated $U(1)$ theories by combining phase-flip codes with Gauss-law checks; Spagnoli, Roggero, and Wiebe extend this to gauge-covariant codes for full fault-tolerance [?] and to native qudit stabiliser codes on $\mathbb{Z}_N$ lattice gauge theories with dynamical matter [?]; Yao [?] and Bradshaw [?] develop QEC codes specific to truncated $SU(2)$ gauge theory (stabiliser and approximate-correction respectively); Lacambra *et al.* [?] identify the gauge-invariant code subspace as a vacuum code in arbitrary Abelian gauge theories; and Rothlin *et al.* [?] use quantum reference frames to show lattice QED itself behaves as a non-stabiliser QEC code. At the same time, the resource arithmetic has moved: the qudit triorthogonal distillation family [14, 15] has $\gamma \approx 1.51$, beating every qubit triorthogonal code below ~ 300 physical qudits, and Gottesman’s original prime- d stabilizer construction [16] already spelled out the foundational differences from the qubit case. *Exotic*—nonstabilizer—codes like the Kubischta 2I code [17] and the Denys–Leverrier covariant 2T/2O codes [18, 19] carry transversal non-Clifford groups that sidestep the Anderson–Jochym–O’Connor qubit-stabilizer no-go [20] entirely. The qudit landscape is therefore not just a technical variant of the qubit one; it is where several of the design bottlenecks of qubit fault-tolerance do not apply.

On qubits the standard gate-set choice has a standard resolution. Clifford+ T [2, 21] pairs the Clifford group with $T = \text{diag}(1, e^{i\pi/4})$; Clifford gates are transversal on the Steane, Reed–Muller, and Bravyi–Haah families, and T is distilled via Bravyi–Kitaev [2] or Bravyi–Haah [3]. Ross–Selinger [22] and Kliuchnikov–Maslov–Mosca [23] make single-qubit synthesis into Clifford+ T a solved problem with certified T -count bounds. For $d = 2$ the choice of gate set is effectively fixed.

No comparable consensus exists at $d > 2$. Howard and Vala [24] proposed a three-parameter family $HV(z, \gamma, \varepsilon)$ of qudit T -gates at prime d , but no systematic comparison of

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members has been done. The qudit finite-subgroup landscape is known [25–29], transversal-gate classifications for stabilizer codes have been cataloged [20], Campbell–Anwar–Browne give prime- d distillation protocols [14], Bocharov, Roetteler, and Svore [30] studied qutrit synthesis for Shor-class algorithms, and Gustafson *et al.* [31] recently extended Ross–Selinger-style exact synthesis to Clifford+ R at $d = 3$. What is missing is a quantitative ranking: given a base group $\mathcal{C}$ and an extension g , how efficiently does $(\mathcal{C}, g)$ synthesize target unitaries?

Slowik, Dulián, and Sawicki [32] supplied the right figure of merit. The *Quantum Circuit Overhead* $Q_T(\mathcal{C}, g)$ is the asymptotic exponent in the word length of an ε -accurate approximation: the number of gates scales as $(1/\varepsilon)^{Q_T}$, bounded below by the Kesten–McKay optimum Q_{opt} and saturated by the “golden gates” of Sarnak [33].

This paper uses Q_T to rank candidates. We introduce SWIFTBOT, an agentic workflow that enumerates $(\mathcal{C}, g)$ pairs at $d = 2, 3, 4$, screens each for universality via Sawicki’s criterion, evaluates Q_T numerically [32], and cross-references surviving pairs against a curated fault-tolerance catalog. With it we make three contributions. First, at $d = 2$ we reproduce the Slowik *et al.* numerics at the percent level (Section IV). Second, we establish the first paper-scale Q_T ranking at $d = 3$ across the three irreducible-adjoint Σ -series subgroups with $|\mathcal{C}| \in \{216, 648, 1080\}$, using Howard–Vala, rational-phase, and Haar-random extensions (Section V). Third, applied to a gate set relevant for discrete-subgroup lattice gauge simulation [7, 11, 12], a non-canonical rational-phase extension beats Clifford+ T by 6.8 $\times$ in mean synthesis depth (Section VI)—showing that the best general-purpose extension and the best target-specific extension can differ by nearly an order of magnitude, a distinction invisible when Q_T alone is reported.

Code and data are released with this paper [34]. Section II reviews the QCO framework, Section III sketches the pipeline, Sections IV to VI present the results, and Section VII closes with a fault-tolerance audit and open problems.

II. BACKGROUND

We briefly review the three ingredients used throughout: the Quantum Circuit Overhead Q_T [32] as our primary figure of merit, the Sawicki criterion [35] for ruling out non-universal extensions without running QCO, and the Howard–Vala family [24] and DSA primitive-gate series [11, 12] as the principal sources of candidate extensions at $d \geq 3$.

II.1. δ -approximate t -designs and Q_T

Let $\mathcal{S} = \mathcal{C} \cup \mathcal{C}^{-1} \cup \{g, g^{-1}\} \subset \text{SU}(d)$ be a symmetric generating set consisting of a finite subgroup $\mathcal{C}$ and a single extension g . Consider the t -design operator on the

t -th tensor power of the defining representation,

$$T_t(\mathcal{S}) = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \pi_t(s), \quad \pi_t = \bigoplus_{\lambda \in \Lambda_t} \pi_\lambda^{\oplus m_\lambda}, \quad (1)$$

where Λ_t is the set of $\text{SU}(d)$ highest weights that appear in the decomposition of $(\mathbb{C}^d)^{\otimes t} \otimes (\mathbb{C}^d)^{* \otimes t}$ and π_λ is the associated irreducible representation. The t -design spectral radius is

$$\delta_t(\mathcal{S}) = \max_{\lambda \in \Lambda_t \setminus \{0\}} \|T_t(\mathcal{S})\|_{\pi_\lambda}, \quad (2)$$

the largest spectral norm across non-trivial irreps. A smaller δ_t means the random walk generated by $\mathcal{S}$ equidistributes faster on $\text{SU}(d)$.

For an ε -net of $\text{SU}(d)$ in the operator norm, Slowik *et al.* [32] show that the number of Solovay–Kitaev-style words required scales as $(1/\varepsilon)^{Q_T}$, with

$$Q_T(\mathcal{S}) = \frac{\log |\mathcal{C}|}{\log(\delta_t^{-1}(\mathcal{S}))}, \quad (3)$$

in the limit $t \rightarrow \infty$. The Kesten–McKay bound for a k -regular Cayley graph gives

$$\delta_t \geq \delta_{\text{opt}} = \frac{2\sqrt{|\mathcal{C}| - 1}}{|\mathcal{C}|}, \quad (4)$$

and therefore

$$Q_T \geq Q_{\text{opt}} = \frac{\log |\mathcal{C}|}{\log(1/\delta_{\text{opt}})}, \quad (5)$$

with equality for “golden gates” [33]. Representative values: $Q_{\text{opt}} \approx 3.47$ for the qubit Clifford group, 4.19 for the projective $A_4 \subset \text{PU}(2)$ (registered as *hurwitz*), 2.81 for the binary icosahedral group $2I$ (BI), and 2.69 for the smallest $d = 3$ Σ -series group $\Sigma(72 \times 3)$.

In practice Q_T is estimated at finite t by sampling the ensemble in Eq. (1) with a Monte-Carlo extension gate g (the *rnd* baseline) or a deterministic one, then evaluating Eq. (2) via sparse spectral decomposition on each irrep of π_t . The feasibility ceiling at $d = 3$ is set by representation dimensions [32]: the largest irrep at $t = 50$ has dimension $\sim 10^5$, whereas $t = 15$ peaks at $\sim 4 \times 10^3$ and is tractable. Throughout this work we take t large enough that the estimate of Q_T has stabilised within a few percent, and we quote Q_{opt} alongside Q_T to expose the gap to optimality.

II.2. Pre-screening with the Sawicki criterion

Not every candidate extension needs a QCO run. Sawicki and Karnas [35] proved that $\mathcal{S}$ generates a dense subgroup of $\text{SU}(d)$ if and only if

- (i) the adjoint representation of $\mathcal{S}$ is irreducible, and
- (ii) $\mathcal{S}$ is not contained in a maximal finite subgroup of the centralizer.

Let $\text{Comm}(\text{Ad}_c)$ denote the space of complex matrices that commute with every Ad_c for $c \in \mathcal{C}$; its dimension is 1 if and only if the adjoint representation of $\mathcal{C}$ is irreducible. Condition (i) is then checkable in $\mathcal{O}(d^2)$ memory via Schur’s orthogonality relation,

$$\dim \text{Comm}(\text{Ad}_c) = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} |\chi_{\text{Ad}}(c)|^2, \quad (6)$$

where $\chi_{\text{Ad}}(c) = \text{tr Ad}_c$. A value > 1 means the adjoint splits, and no extension preserving that splitting can be universal. This immediately excludes from consideration, e.g., $\Sigma(36 \times 3)$ (the 108-element crystallographic group), the 60-element simple group $\Sigma(60) \simeq A_5$, and the $s_{60 \times 4}$ family at $d = 3$, since any diagonal extension preserves the reducible-adjoint structure. Condition (ii) is enforced at finite cost by running a bounded closure $\langle \mathcal{S} \rangle \cap \text{Cap}(n)$ against the classification of finite subgroups of $\text{SU}(d)$ (orders ≤ 120 at $d = 2$, ≤ 1080 at $d = 3$, ≤ 5040 at $d = 4$) [25–29]; we use cap sizes $n = 120, 1080, 5040$ respectively. An extension is flagged `finite` if closure terminates below the cap, `universal_likely` if closure survives the cap and condition (i) holds, and `infinite.but_not_dense` if condition (i) fails but closure is open.

II.3. Candidate extensions

At $d = 2$ the canonical extension is $T = P(\pi/4) = \text{diag}(1, e^{i\pi/4})$, giving the Clifford+ T set with certified synthesis via Ross–Selinger [22] and Kliuchnikov–Maslov–Mosca [23]. We also consider diagonal $P(\theta)$ for $\theta \in \{\pi/8, 2\pi/9, \pi/9, \pi/18\}$, the Parzanchevski–Sarnak supergolden gates [36], and Haar-random extensions.

At prime d the Howard–Vala family [24] generalizes T via three integer parameters (z, γ, ε) :

$$\text{HV}(z, \gamma, \varepsilon) = \mathcal{N} \cdot \text{diag}(\zeta_d^{e_0}, \zeta_d^{e_1}, \dots, \zeta_d^{e_{d-1}}), \quad (7)$$

with $\zeta_d = e^{2\pi i/d}$, exponents $e_k = z k^3 + 3\gamma k^2 + \varepsilon$, and $\mathcal{N}$ a phase chosen so that $\det \text{HV}(z, \gamma, \varepsilon) = 1$. Our panel uses $(z, \gamma, \varepsilon) \in \{(1, 1, 0), (1, 2, 0), (2, 1, 0)\}$: the first is Campbell’s magic gate [14], the second is the Howard–Vala Eq. (27) variant (Clifford-equivalent at $d = 3$), and the third is a representative new orbit.

Discrete-subgroup approximations of lattice gauge simulations for $\text{SU}(d)$ [11, 12] also pick out a different kind of target. For each chosen qudit group, the continuous degree of freedom reduces to a trace rotation $U_{\text{Tr}}(\theta)$ that decomposes into a finite set of $R_Z(\alpha_c \theta)$ rotations whose coefficients α_c depend on the specific subgroup and encoding. This motivates a *target-family* figure of merit: the mean breadth-first-search depth to synthesize the required R_Z rotations within $\varepsilon \leq 10^{-2}$. We instantiate it for one specific subgroup in Section VI.

III. SWIFTBOT PIPELINE

SWIFTBOT is a five-stage agentic pipeline built around the QCO kernel of [32]. Numerical stages are deterministic; an LLM supervisor intervenes only at stage boundaries to rank candidates, propose extensions, and decide when to stop. All headline numbers below come from deterministic panels that bypass the LLM; the agent is used for exploratory search not reported here. Full code and a command-line driver are released with the paper [34].

Stage 1 (group registration) computes projective closures $\langle \mathcal{G} \rangle \subset \text{SU}(d)/Z(\text{SU}(d))$ from seed unitaries and stores each group under a content-addressable key. The registry covers the canonical $d \in \{2, 3, 4\}$ subgroups used here: Pauli–Clifford, the projective $A_4 \subset \text{PU}(2)$ (registered as `hurwitz`), and $2I$ (BI) at $d = 2$; the $\Sigma(m \times 3)$ series for $m \in \{36, 72, 216, 360\}$ at $d = 3$; and selected Fairbairn subgroups at $d = 4$.

Stage 2 (universality screen) evaluates $\dim \text{Comm}(\text{Ad}_c)$ via Eq. (6) and runs a bounded closure against the classification caps $\{120, 1080, 5040\}$ [25–29]. Pairs whose closure terminates below cap are tagged `finite` and skipped; reducible-adjoint pairs are tagged diagnostically.

Stage 3 (QCO numerics) invokes the kernel of [32] as a subprocess. Two optimizations help: (i) every gate has the form $g c g^\dagger$ with $c \in \mathcal{C}$, so $\pi_\lambda(g c g^\dagger) = \pi_\lambda(g) \pi_\lambda(c) \pi_\lambda(g)^\dagger$ replaces $|\mathcal{C}|$ exponentials with one; (ii) the $\pi_\lambda(c)$ lists are persisted to disk keyed by a fingerprint of $\mathcal{C}$ and the highest weight λ . Together these give 3–6 $\times$ speedup at $d = 3$. Results are stored as per-sample `QTRecords` in a SQLite cache with full provenance.

Stages 4–5 (fault-tolerance) match each surviving $(\mathcal{C}, g)$ against a curated catalog of QEC codes (Steane, Reed–Muller, Bravyi–Haah, Kubischta $2I$, qutrit triorthogonal, ternary Golay, Pollatsek–Ruskai PI, Kubischta–Teixeira PI, and more) and distillation protocols (Bravyi–Kitaev, Bravyi–Haah, Campbell–Anwar–Browne, programmable MSD, OJB code-switching). Unmatched pairs return a *research-needed* marker with an EC Zoo [37] pointer rather than guessing.

IV. VALIDATION AT $d = 2$

We validate the SWIFTBOT QCO kernel against the $d = 2$ reference dataset of Słowiak, Dulián, and Sawicki [32] on two qubit bases: the Clifford group ($|\mathcal{C}| = 24$) and the 12-element projective $A_4 \subset \text{PU}(2)$ (registered `hurwitz`), each paired with $P(\pi/4)$ and with Haar-random extensions. We also run a paper-scale 10^4 -sample Haar search on the binary icosahedral group $2I$ (BI, $|\mathcal{C}| = 120$), which carries the Kubischta nonadditive transversal code [17]. All runs are at $t = 50$, matching the reference. A zeroth-order check—replaying the δ -parser on all 5050 rows of the reference file and recomputing Q_T —matches the reference to $\mathcal{O}(10^{-15})$, isolating any later disagreement to the spectral kernel. Headline results are summarized in Table I: Clifford+ $P(\pi/4)$ yields $Q_T = 51.9$ (reference

≈ 52); Haar-random extensions agree at the few-percent level.

Two remarks. First, Clifford+ T is about $14\times$ worse than Haar-random on Q_T (51.9 vs 3.79)—commensurability between $P(\pi/4)$ and the Clifford orbit slows the Haar random walk by more than an order of magnitude in the approximation exponent [32]. This is a Q_T claim, not a verdict on Clifford+ T itself: when the target is structured (Section VI), the ranking inverts. Second, the BI ratio $Q_T/Q_{\text{opt}} \approx 1.27$ at 10^4 samples is far closer to saturation than Clifford’s and arrives with a cataloged fault-tolerant code [17]; the empirical δ distribution on BI has median 0.38 ($Q_T \simeq 4.97$) and 5th-percentile 0.31, so the best-of- 10^4 floor is the natural super-golden candidate predicted by [33].

V. $d = 3$ SURVEY

With the SWIFTBOT kernel validated at $d = 2$ (Section IV), we turn to the first paper-scale Q_T ranking at $d = 3$. The survey covers the three irreducible-adjoint Σ -series subgroups of $\text{SU}(3)$ — $\Sigma(72 \times 3)$, $\Sigma(216 \times 3)$, and $\Sigma(360 \times 3)$ —each extended by a common panel of eight candidate gates. The smaller groups $\Sigma(36 \times 3)$ (the 108-element crystallographic Valentiner analogue) and $\Sigma(60) \simeq A_5$ have reducible adjoint representations (cf. Eq. (6)) and are excluded at Stage 2 of the pipeline: no extension preserving the invariant subspace structure of $\text{su}(3)$ can render them universal, so their Q_T is formally divergent [35].

V.1. Panel specification

The panel fixes eight extensions per base group, with identical parameters across the three groups. Three are Howard–Vala variants $\text{HV}(1, 1, 0)$, $\text{HV}(1, 2, 0)$, $\text{HV}(2, 1, 0)$ (see Eq. (7)). Three are diagonal phase gates $P(\theta)$ at angles $\theta \in \{2\pi/9, \pi/9, 2\pi/5\}$, chosen to span both the cyclotomic-9th-root sector relevant to Section VI and the coarser $2\pi/5$ angle that generates no low-order closure against any of the three base groups. The remaining two slots per group are independent Haar-random batches of size 10, drawn freshly per subprocess from the invariant measure on $\text{SU}(3)$; these serve as the *rnd* baseline of [32]. All eight extensions survive the Stage 2 universality screen of Section II.2: each Σ -series base has one-dimensional adjoint commutant by Eq. (6), and the bounded closures stay open past the $|\mathcal{C}| \leq 1080$ cap of [25–29].

The QCO numerics run at $t = 5$, the largest design order for which the full panel completes on a single 32-core workstation within the budget of an interactive session. Fixed extensions use `sample_size = 1` (the gate is deterministic); Haar-random batches use `sample_size = 10`. Each (group, extension) pair is dispatched as an independent `qco` subprocess with `OMP_NUM_THREADS=1` and `--workers 8` to keep the per-irrep spectral evaluation of

Eq. (2) single-threaded while exploiting thread-level parallelism across the $|\mathcal{C}|$ matrix actions described in Section III. The full Tier-1 panel—3 groups $\times$ 8 extensions, with the cached $\pi_\lambda(\mathcal{C})$ tables rebuilt once per group—completes in approximately 17 minutes of wall time. All numerical values reported below are reproduced bit-identically from the SQLite cache released with the paper [34].

V.2. Tier-1 ranking

Table II lists every (group, extension) pair at $t = 10$, ranked within each group by δ . The $Q_T/\log|\mathcal{C}|$ column divides out the base-group scale and exposes the spectral-gap content of the extension.

Fig. 1 shows three features, elaborated against Table II. (i) *Structured-extension degeneracy*. Five of six structured extensions give δ identical to four decimals across all three base groups, bit-for-bit the same as at $t = 5$. $\text{HV}(2, 1, 0)$ is the only exception: it lifts from the $t = 5$ floor $\delta = 0.5000$ on the smaller groups to $\delta \approx 0.58$ at $t = 10$, while $\Sigma(360 \times 3)$ stays at 0.5698. The degeneracy is structural, not a finite- t artifact. (ii) *Linear Q_T growth with $\log|\mathcal{C}|$* . Because δ is essentially constant, Q_T grows linearly with $\log|\mathcal{C}|$ for the degenerate extensions: a larger base group needs proportionally more depth to densely cover $\text{SU}(3)$, so the normalized $Q_T/\log|\mathcal{C}|$ (rightmost column) collapses the three groups onto the same value. (iii) *Haar-random gains with $|\mathcal{C}|$* . Best-sample Haar δ drops from 0.26 to 0.18 from $\Sigma(72 \times 3)$ to $\Sigma(360 \times 3)$; a random draw couples to every Lie-algebra direction including the subdominant channels that grow with $|\mathcal{C}|$. Statistically significant Haar distributions are deferred to Sections V.3 and V.4.

V.3. Apples-to-apples Haar panel

To separate the base-group effect from random-extension variance, we generated a single batch of ten Haar-random single-qutrit extensions (`seed=42`) and evaluated the panel identically on all three groups. Table III lists the best-sample δ per matrix.

Seven of the ten matrices have δ on $\Sigma(360 \times 3)$ strictly smaller than δ on $\Sigma(72 \times 3)$, confirming that a typical Haar extension is improved by the larger base group. The per-matrix ordering against $\Sigma(216 \times 3)$ is noisier: four of ten matrices have δ on $\Sigma(360 \times 3)$ smaller than on $\Sigma(216 \times 3)$, and the ten-matrix apples-panel mean δ is marginally worse on $\Sigma(360 \times 3)$ (0.2550) than on $\Sigma(216 \times 3)$ (0.2506). The culprit is a single outlier: `apple05` gives $\delta = 0.4377$ on $\Sigma(360 \times 3)$ against 0.2446 on $\Sigma(72 \times 3)$ and 0.2412 on $\Sigma(216 \times 3)$, inflating the $\Sigma(360 \times 3)$ mean by roughly one standard deviation of the rest of the panel.

A paper-scale rerun at $N = 100$ matrices (same `seed = 42`; the first ten coincide with Table III) resolves the ambiguity. The three panel means land at $\mu_{216} = 0.3214$, $\mu_{648} = 0.2623$, $\mu_{1080} = 0.2118$ with standard deviations 0.073, 0.074, and 0.090 respectively (Fig. 2). The $\Sigma(360 \times 3)$

TABLE I. $d = 2$ validation of the SWIFTBOT QCO kernel against the reference dataset of Słowik, Dulián, and Sawicki [32]. All values computed at $t = 50$; the Haar-random (rnd) rows report the best Q_T across the stated number of samples drawn from the invariant measure on $SU(2)$; the fixed-diagonal row is deterministic. The Q_{opt} column is the Kesten–McKay floor of Eq. (5). Reference values are the $t \rightarrow \infty$ headline numbers of [32] when available; BI has no published QCO reference.

Base ($ \mathcal{C} $)	Extension	Q_T (this work)	Q_T [32]	Q_{opt}
Clifford (24)	$P(\pi/4)$	51.91	≈ 52	3.47
Clifford (24)	rnd (best of 20)	3.79	≈ 3.7	3.47
hurwitz (12)	rnd (best of 20)	4.34	≈ 4.0	4.19
BI (120)	rnd (best of 10^4)	3.57	—	2.81

TABLE II. $d = 3$ survey at $t = 10$. Each fixed extension is deterministic (δ from a single evaluation); the rnd row per group reports the best of two independent Haar draws, each itself a single-sample evaluation at this design order. Kesten–McKay floor Q_{opt} : 2.692, 2.544, 2.495 for $\Sigma(72, 216, 360 \times 3)$. Rows are ranked within each group by δ ; all structured extensions match their $t = 5$ values to four decimals except HV(2, 1, 0) on the two smaller groups, where the $t = 5$ floor at $\delta = 0.5000$ lifts at $t = 10$.

Group	Extension	kind	δ	Q_T	$Q_T / \log \mathcal{C} $
$\Sigma(72 \times 3)$	rnd (best of 2)	rnd	0.2605	3.996	0.744
	HV(2, 1, 0)	howard_vala	0.5927	10.277	1.911
	$P(2\pi/5)$	angle	0.6545	12.681	2.360
	HV(1, 1, 0)	howard_vala	0.6764	13.750	2.559
	HV(1, 2, 0)	howard_vala	0.6764	13.750	2.559
	$P(2\pi/9)$	angle	0.8830	43.208	8.041
	$P(\pi/9)$	angle	0.9698	175.561	32.671
$\Sigma(216 \times 3)$	rnd (best of 2)	rnd	0.2107	4.157	0.643
	HV(2, 1, 0)	howard_vala	0.5833	12.011	1.856
	$P(2\pi/5)$	angle	0.6545	15.273	2.360
	HV(1, 1, 0)	howard_vala	0.6764	16.561	2.559
	HV(1, 2, 0)	howard_vala	0.6764	16.561	2.559
	$P(2\pi/9)$	angle	0.8830	52.039	8.041
	$P(\pi/9)$	angle	0.9698	211.443	32.671
$\Sigma(360 \times 3)$	rnd (best of 2)	rnd	0.1784	4.052	0.579
	HV(2, 1, 0)	howard_vala	0.5698	12.419	1.775
	$P(2\pi/5)$	angle	0.6545	16.478	2.360
	HV(1, 1, 0)	howard_vala	0.6764	17.867	2.559
	HV(1, 2, 0)	howard_vala	0.6764	17.867	2.559
	$P(2\pi/9)$	angle	0.8830	56.145	8.041
	$P(\pi/9)$	angle	0.9698	228.127	32.671

mean now sits strictly below the $\Sigma(216 \times 3)$ mean by $\mu_{648} - \mu_{1080} = 0.051$, a separation of roughly 5.7 standard errors of the mean. The ten-matrix inversion reported above is therefore confirmed to be Monte-Carlo noise from the single **apple05** outlier; across a Haar ensemble at 100-matrix statistics the expected δ *does* decrease monotonically with $|\mathcal{C}|$ on the irreducible-adjoint Σ -series, strengthening Finding 3.

V.4. Random-diagonal ternary scan

The Tier-1 and Tier-2 panels show a striking feature: five of six structured diagonal and Howard–Vala extensions give δ values identical to four decimals across $\Sigma(72 \times 3)$ and $\Sigma(216 \times 3)$. A natural question is whether this four-decimal degeneracy is specific to the cyclotomic content of the panel’s three phases $P(2\pi/9)$, $P(\pi/9)$, $P(2\pi/5)$, or

whether it applies to the full maximal torus of $SU(3)$. We answer by scanning $N = 50$ random $SU(3)$ diagonal extensions $\text{diag}(e^{i\theta_1}, e^{i\theta_2}, e^{i\theta_3})$ (**seed** = 42, θ_1, θ_2 uniform on $[0, 2\pi)$, $\theta_3 = -\theta_1 - \theta_2 \bmod 2\pi$) on all three base groups at $t = 5$, a $3 \times 50 = 150$ -evaluation panel completed in ~ 14 minutes of wall time on a 32-core node.

The result is stronger than the structured-panel observation: for every one of the 50 random diagonals, the δ value on $\Sigma(72 \times 3)$ and $\Sigma(216 \times 3)$ agrees *bit-for-bit*, while $\Sigma(360 \times 3)$ is generically distinct. Per-group summary statistics (best / mean / max) across the 50 samples are

$$\Sigma(72 \times 3), \Sigma(216 \times 3) : 0.4516 / 0.6769 / 0.9921, \quad (8)$$

$$\Sigma(360 \times 3) : 0.2539 / 0.5106 / 0.9662. \quad (9)$$

Fig. 3 plots the full distribution on the $SU(3)$ maximal-torus simplex; the Σ -series base groups have a permutation subgroup (the qutrit Clifford contains $SU(3)$ Weyl elements), so each random-diagonal point is S_3 -symmetrized

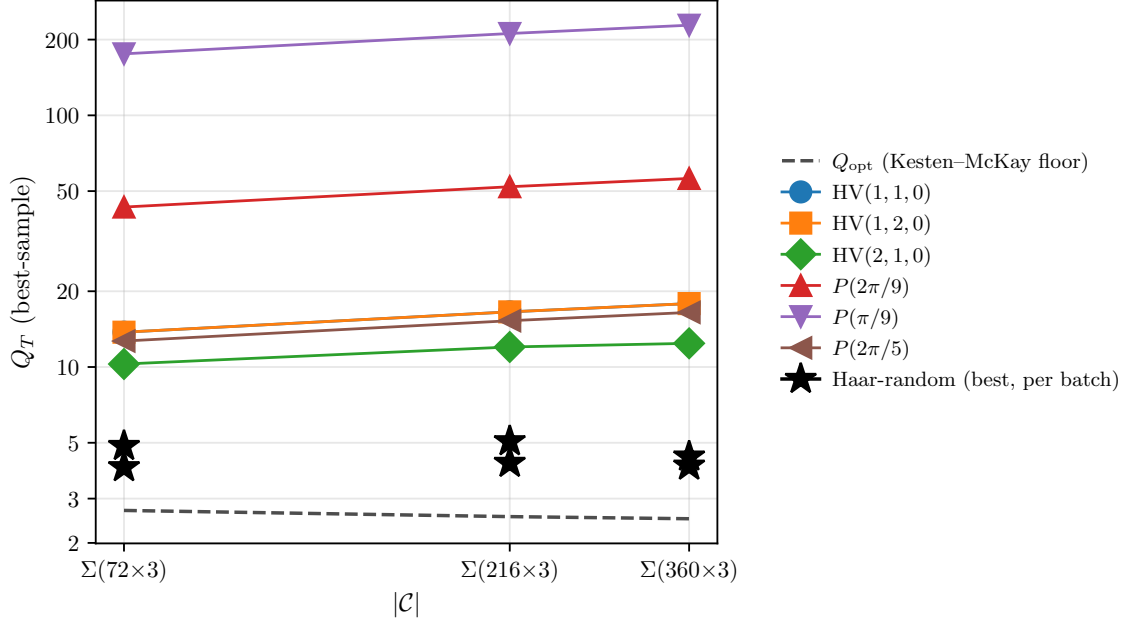


FIG. 1. Q_T across the three Σ -series base groups at $d = 3$, $t = 10$, for every extension in the panel of Table II. The dashed line is the Kesten–McKay floor Q_{opt} of Eq. (5); rational-angle diagonal and Howard–Vala extensions run nearly parallel to it at a factor ~ 3 –70 offset (Finding 2), while Haar-random extensions (stars) approach Q_{opt} as $|C|$ grows (Finding 3).

TABLE III. Apples-to-apples Haar panel at $t = 5$: ten pre-generated single-qutrit extensions (`seed=42`) evaluated identically against each of the three base groups. The smallest δ in each row is bolded. Mean δ over the ten matrices appears in the last row; lower is better.

matrix	δ		
	$\Sigma(72 \times 3)$	$\Sigma(216 \times 3)$	$\Sigma(360 \times 3)$
apple00	0.2974	0.2974	0.3462
apple01	0.4636	0.4636	0.2029
apple02	0.2221	0.1727	0.2285
apple03	0.2493	0.2292	0.2299
apple04	0.3430	0.2104	0.1197
apple05	0.2446	0.2412	0.4377
apple06	0.2842	0.1701	0.1965
apple07	0.3169	0.2312	0.2271
apple08	0.3996	0.2377	0.3795
apple09	0.2526	0.2526	0.1823
mean	0.3073	0.2506	0.2550

to six positions with the same δ , exposing the full torus symmetry.

The Finding-1 degeneracy is therefore *not* a property of the cyclotomic angles we chose to probe but a structural property of the nested-subgroup relation $\Sigma(72 \times 3) \subset \Sigma(216 \times 3)$: at $t = 5$, the dominant t -design irreps of $\text{SU}(3)$ restrict to representations that both groups act on identically, so any diagonal extension couples through the same spectral channel on both. $\Sigma(360 \times 3)$ does not sit in this chain, hence its breakout. A $t = 10$ rerun would

test whether a subdominant irrep of $\Sigma(216 \times 3)$ that is absent in $\Sigma(72 \times 3)$ becomes spectrally visible and lifts the degeneracy.

V.5. Best achievable Q_T

Combining the two panels, the best (group, extension) pair is **apple04** on $\Sigma(360 \times 3)$ at $t = 5$ with $\delta = 0.1197$ and $Q_T = 3.291$ —within 32% of $Q_{\text{opt}} = 2.495$. At $t = 10$ the best fixed extension is $\text{HV}(2, 1, 0)$ on $\Sigma(72 \times 3)$, with $Q_T = 10.28$, a factor 3.8 above $Q_{\text{opt}} = 2.692$. The shape of the ranking is the same at both t : Haar-random extensions dominate every structured extension, $\text{HV}(2, 1, 0)$ leads the structured field, and the three diagonal-phase angles $P(2\pi/9)$, $P(\pi/9)$, $P(2\pi/5)$ all give worse Q_T than even the worst Howard–Vala variant. The narrow-angle $P(\pi/9)$ in particular produces δ within 3% of unity, inflating Q_T by two orders of magnitude relative to the Haar baseline—a direct analogue of the $14 \times$ Clifford+ T penalty at $d = 2$ (Section IV).

The $t = 5$ and $t = 10$ data agree on the Tier-1 panel to four decimals for all structured extensions except the $\text{HV}(2, 1, 0)$ floor lift noted above. The four-decimal cross-group δ -degeneracy is therefore not a finite- t artifact; it is a structural feature of rational-angle diagonal and Howard–Vala extensions on the Σ -series. Rnd batches at $t = 10$ are single-sample draws (the qco optimized kernel aggregates to one row per subprocess regardless of ss); statistically significant Haar statistics come from the apples-100 panel in Section V.3 and from the 10^4 -sample

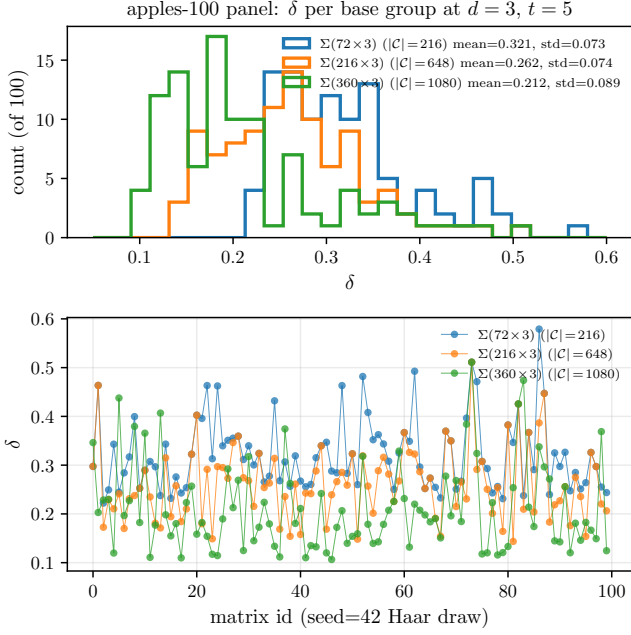


FIG. 2. Apples-to-apples Haar panel at $N = 100$ matrices ($\text{seed} = 42$) on the three $d = 3$ Σ -series base groups. Top: overlaid δ histograms. Bottom: per-matrix δ for each group; matrix id 0–9 coincides with the ten-matrix subset of Table III. The monotone reduction $\mu_{216} > \mu_{648} > \mu_{1080}$ predicted by Finding 3 is now resolved at $\sim 5.7\sigma/\sqrt{N}$; the single `apple05` outlier visible as the $\Sigma(360 \times 3)$ peak near $\delta \simeq 0.44$ is the one that inverted the ten-matrix ordering in Table III.

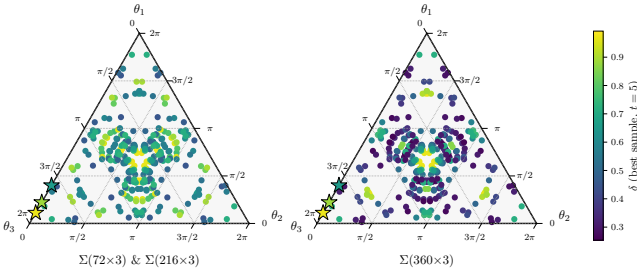


FIG. 3. δ on the $SU(3)$ maximal torus at $t = 5$. Each panel shows the 50 random-diagonal extensions of Section V.4, S_3 -symmetrized to six copies each (300 points per panel). The left panel, labeled $\Sigma(72 \times 3)$ & $\Sigma(216 \times 3)$, reports the δ values that are bit-for-bit identical on the two smaller Σ -series groups; the right panel reports $\Sigma(360 \times 3)$. Red stars mark the three structured rational-angle extensions $P(2\pi/9)$, $P(\pi/9)$, $P(2\pi/5)$ from Table II, colored by their own δ on each group; they sit on the $\theta_1 = 0$ edge because each has the form $\text{diag}(1, e^{i\theta}, e^{-i\theta})$.

BI pilot at $d = 2$ (Section IV).

VI. THE DSA PRIMITIVE-GATE TARGET FAMILY

The $d = 3$ survey of Section V ranks extensions by the Haar-uniform figure of merit Q_T . Many applications do not need a uniform net. Discrete-subgroup approximations of lattice gauge simulations for $SU(d)$ [11, 12] require instead a small structured family of unitaries: a few continuous-angle R_Z rotations with group-theoretic coefficients, plus a finite set of cyclotomic phases from the discrete Fourier transform U_F . On such a workload the relevant figure of merit is the mean synthesis cost over that finite family, not a t -design spectral gap. The two figures of merit disagree in general.

A concrete target family. We work out the case of the $\Sigma(36 \times 3)$ encoding as a representative example; different subgroups in the series produce different coefficient sets. For $\Sigma(36 \times 3)$, the trace gate $U_{\text{Tr}}(\theta) = \text{diag}_{g \in C}(e^{i\theta \text{Re Tr}(g)})$, after the three-qubit encoding, reduces to five continuous $R_Z(\alpha_c \theta)$ rotations with $\alpha_c \in \{1, 3, 5, 9, 15\}$. These coefficients are specific to $\Sigma(36 \times 3)$; other subgroups in the series produce different trace spectra and different α_c . The discrete half comes from U_F : up to global phase, the Howard–Vala qutrit gate $HV(1, 2, 0)$ [24] injects 9th-root phases, requiring $R_Z(2\pi k/9)$ for $k = 1, \dots, 8$. The synthesis target for the $\Sigma(36 \times 3)$ qubit pipeline is therefore

$$\mathcal{T}_{36} = \{R_Z(2\pi k/9)\}_{k=1}^8 \cup \{R_Z(\alpha_c \theta)\}_{\alpha_c \in \{1, 3, 5, 9, 15\}, \theta \in (0, \pi)} \quad (10)$$

Mean-depth figure of merit. Given a qubit base group $\mathcal{C}$ and single-gate extension g , SWIFTBOT enumerates the word tree of $\mathcal{C} \cup \{g, g^{-1}\}$ by breadth-first search with projective-phase deduplication, capped at 2×10^5 unique words. For each $U \in \mathcal{T}_{36}$ we record the shortest BFS word whose projective distance $d(U, W) = \sqrt{2 - |\text{Tr}(U^\dagger W)|}$ falls below $\varepsilon = 10^{-2}$, or the depth of closest approach if no such word exists. Discrete targets may be hit *exactly* at finite depth, in which case the corresponding distance is zero and the T -count is read directly from the word. Continuous targets are sampled at ten uniform $\theta \in (0, \pi)$ per α_c , for fifty continuous evaluations per extension. We report mean distance across Eq. (10) as the single headline number; ties are broken on the discrete-exact count.

Result. Table IV ranks the candidates at BFS depth 15, $\varepsilon = 10^{-2}$, $N_\theta = 10$ per continuous coefficient. The Clifford+ $P(2\pi/9)$ extension attains mean projective distance 0.0122 on $\mathcal{T}_{36}$ with 27 of 58 targets hit below ε , while the canonical Clifford+ T extension attains mean distance 0.0833 with only 5 hits—a factor 6.8 $\times$ separation in mean distance. The equivalence class of $P(2\pi k/18)$ with $\text{gcd}(k, 18) = 1$ (which includes $P(\pi/9)$ and $P(\pi/18)$) all saturate the discrete 9th-root targets, a direct consequence of $\langle \text{Clifford}, P(2\pi/9) \rangle$ closing at a finite group of $\sim 3 \times 10^4$ elements; the algebraic ring $\mathbb{Z}[\omega_{18}, \frac{1}{2}]$ supplies every 9th-root phase via a short word. Within this equivalence class, Clifford+ $P(\pi/18)$ marginally edges Clifford+ $P(2\pi/9)$ at depth 15 (mean 0.0104 vs. 0.0122; 34 vs. 27 hits), reflecting the strictly larger 18th-root cover-

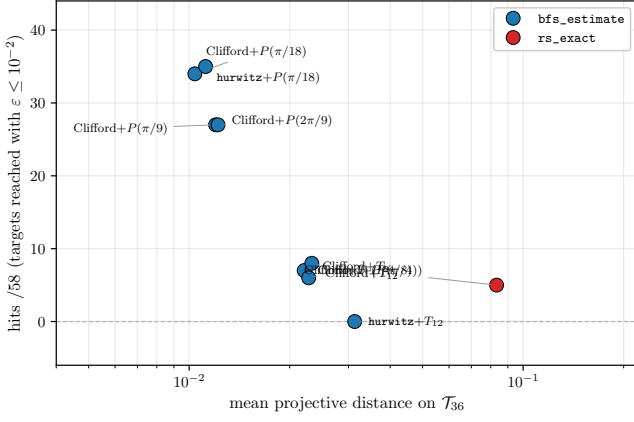


FIG. 4. DSA $\Sigma(36 \times 3)$ target-family coverage from Table IV: hits/58 versus mean projective distance on $\mathcal{T}_{36}$ at BFS depth 15, $\varepsilon = 10^{-2}$. Canonical Clifford+ T (red, **rs_exact** certified) sits bottom-right with the largest mean distance and only 5 hits. Rational-angle $P(k\pi/18)$ extensions (blue, **bfs_estimate**) cluster top-left with many hits at an order of magnitude smaller distance; super-golden T_{12}/T_{24} rows fall in between despite saturating Q_{opt} (Section IV). The visual separation matches the 6.8 $\times$ headline.

age: $P(\pi/18)$ generates ω_{36} , which contains both the 9th- and 18th-root phases of U_F . Crucially, the super-golden extensions T_{24} and T_{12} —which optimize Q_T for Clifford and for the 12-element projective A_4 base **hurwitz** respectively [33]—perform *worse* on $\mathcal{T}_{36}$ than the rational-angle rows: their algebra $\mathbb{Z}[i, \sqrt{2}]$ does not contain ω_9 , so they cannot reach the 9th-root targets exactly at any depth; on the $|\mathcal{C}| = 12$ **hurwitz** base the T_{12} extension hits zero targets at depth 15. The **hurwitz**+ $P(\pi/18)$ row is competitive (35 hits, mean 0.0112) and does so with a strictly *smaller* base group: in the target-family regime the overhead advantage tracks the cyclotomic content of the extension, not the size of the base.

Interpretation. Clifford+ $P(2\pi/9)$ is worse on Q_T than Clifford+ T yet 6.8 $\times$ better on mean depth over $\mathcal{T}_{36}$. The reason is structural: Q_T is the coefficient of a Haar-uniform covering rate, while $\mathcal{T}_{36}$ is a measure-zero algebraic slice of $\text{SU}(2)$ whose ω_9 cyclotomic content matches $\langle \text{Clifford}, P(2\pi/9) \rangle$. The super-golden extensions saturate Q_{opt} but equidistribute in the wrong direction for this target. A lattice-gauge simulator compiling Eq. (10) should therefore rank extensions by target-family mean depth, not by Q_T , and the two rankings can disagree by nearly an order of magnitude.

Caveats. The Clifford+ T row’s T -count is certified by Ross–Selinger exact synthesis [22, 23]. No analogous normal form exists for $\mathbb{Z}[\omega_{18}, \frac{1}{2}]$, so other rows carry **bfs_estimate** upper bounds; the 6.8 $\times$ advantage is large enough to survive any plausible tightening on the canonical side but is contingent on the broader fault-tolerance stack. Anderson–Jochym–O’Connor [20] excludes direct stabilizer-code distillation of $|k\pi/9\rangle$, but programmable MSD on a Clifford+ T substrate [38–40] and non-stabilizer

code-switching [41, 42] both bypass that no-go. The advantage is cleanest when $P(2\pi/9)$ is a native primitive—e.g., in a qutrit register or on a hypothetical nonadditive code that distills $|\omega_9\rangle$ directly.

VII. DISCUSSION

Two findings drive the discussion below. First, at $d = 3$ and $t = 5$, Haar-random extensions on the largest irreducible-adjoint Σ -series base reach $Q_T/Q_{\text{opt}} \approx 1.32$ —essentially the same “golden-gate direction” observed at $d = 2$, where Haar extensions sit 4% above Q_{opt} on the 12-element **hurwitz** base (Section IV). No structured extension in our panel (Howard–Vala or rational diagonal) gets closer than a factor ~ 2.9 of Q_{opt} . Second, and in apparent tension with the first, the non-canonical Clifford+ $P(2\pi/9)$ extension beats the canonical Clifford+ T by 6.8 $\times$ on the DSA primitive-gate target family (Section VI) despite being $\sim 8\times$ worse on Q_T . The two findings together are the paper’s central design lesson: the “best general-purpose extension” and the “best target-specific extension” optimize different objectives, and paper-scale qudit fault-tolerance design must report both.

VII.1. Fault-tolerance status of the winning extensions

A Q_T ranking is necessary but not sufficient for fault-tolerant implementation. Table V audits every extension evaluated against the curated SWIFTBOT catalogs of transversal codes, distillation protocols, and certified synthesis rings. Only canonical Clifford+ T at $d = 2$ admits an end-to-end certified stack and serves as our validation anchor. Three rows inherit a transversal-code half only: **hurwitz**-based super-golden extensions via Parzanchevski–Sarnak [36], BI+rnd via the Kubischta 2I code [17], and the $d = 3$ $\Sigma + \text{HV}(z, \gamma, \varepsilon)$ rows via the qutrit triorthogonal $[[9m - k, k, 2]]_3$ family [14]. All other rows lack either a distillation circuit or a certified synthesis ring.

Three regimes deserve comment. First, the cyclotomic ring $\mathbb{Z}[\omega_{18}, \frac{1}{2}]$ —hosting the $d = 2$ target-family winners $P(2\pi/9)$, $P(\pi/9)$, $P(\pi/18)$ —has no Ross–Selinger-style normal form, and Anderson–Jochym–O’Connor [20] excludes any qubit *stabilizer* code with a transversal $|k\pi/9\rangle$ state. Two non-stabilizer routes bypass the exclusion: programmable MSD on a Clifford+ T substrate [38–40], and measurement-free code-switching between a Clifford stabilizer code and a Kubischta–Teixeira permutation-invariant code [41, 42]. At $d = 3$ the same rotations are qutrit-native and Campbell–Anwar–Browne [14] distills $|\omega_9\rangle$ directly.

Second, the super-golden gates of Parzanchevski–Sarnak [36] live in $\mathbb{Z}[i, \sqrt{2}]$ and are not Clifford-orbit rotations of stabilizer states, so triorthogonal distillation does not apply. Whether a golden-gate-compatible distillation circuit can be built—possibly on the Kubischta

TABLE IV. DSA $\Sigma(36 \times 3)$ target-family coverage at BFS depth 15, $\varepsilon = 10^{-2}$, 10 continuous θ -samples per α_c . The “hits” column counts all 58 targets (the 8 discrete 9th-root phases in Eq. (10) plus 50 continuous $R_Z(\alpha_c \theta)$ samples) reached below ε ; rational-angle rows in the ω_{18} class hit all 8 discrete targets. Mean distance is across all 58 targets; lower is better. $\langle T \rangle_{\text{hit}}$ is the mean T -count along words reaching a hit target. Cost method indicates whether the T -count is certified: Ross–Selinger [22] exact synthesis (**rs_exact**) or SWIFTBOT BFS estimate (**bfs_estimate**). All numbers are reproduced by `swiftbot cover-panel --family lamm_sigma36 --max-depth 15`.

Base + extension	hits/58	mean dist	max dist	$\langle T \rangle_{\text{hit}}$	cost method
Clifford+ $P(\pi/18)$	34	0.0104	0.0420	2.79	bfs_estimate
hurwitz + $P(\pi/18)$	35	0.0112	0.0420	5.37	bfs_estimate
Clifford+ $P(\pi/9)$	27	0.0120	0.0401	2.04	bfs_estimate
Clifford+ $P(2\pi/9)$	27	0.0122	0.0401	2.11	bfs_estimate
Clifford+ $P(\pi/8)$	7	0.0221	0.0387	0.29	bfs_estimate
Clifford+ T_{24}	6	0.0228	0.0465	0.83	bfs_estimate
Clifford+ T_{12}	8	0.0233	0.0474	1.88	bfs_estimate
hurwitz + T_{12}	0	0.0313	0.0573	—	bfs_estimate
Clifford+ $P(\pi/4)$	5	0.0833	0.1554	0.00	rs_exact

2I code, whose transversal group is precisely the super-golden base—is open.

Third, Haar-random extensions are by definition not exactly synthesized, and so cannot be distilled in the usual sense [2, 3]. This applies uniformly to every row; the best current claim is transversal-group support at the base-code level.

For the Σ -series at $d = 3$, data and distillation codes are structurally separable. The Howard–Vala magic state is distilled by the qutrit triorthogonal family [14, 15], but no explicit data code hosts a Σ -series group transversally. Denys–Leverrier covariant-encoding constructions realize $2T$ and $2O$ transversally at $d = 2$ [18, 19]; extending either to a qutrit $\Sigma(m \times 3)$ base—an open case inside the twisted unitary t -group program [41]—would complete the stack. On the synthesis side of the qutrit story, Gustafson *et al.* [31] recently demonstrated Ross–Selinger-analogue exact synthesis of single-qutrit unitaries over Clifford+ R , closing the compilation-certification gap for $d = 3$ Howard–Vala gate sets.

Higher-dimensional qudit MSD is advancing rapidly. Krishna–Tillich [43] opened the low-overhead programme; Wills, Hsieh, and Yamasaki [44] achieved the first constant-overhead MSD construction via asymptotically good quantum codes; Golowich–Guruswami [45] and Nguyen [46] gave asymptotically good qubit codes with transversal non-Clifford and CCZ gates. Most directly applicable to our framework, Cervia *et al.* [47] construct triorthogonal codes on $\mathbb{F}_{2^{2m}}$ at qudit dimensions as small as $q = 64$ that beat the Tsfasman–Vlăduț–Zink bound and give constant-space-overhead $|CCZ\rangle$ distillation; a constant-depth Clifford circuit converts the qudit state down to qubit $|CCZ\rangle$, so the construction is a drop-in upgrade for any Clifford+CCZ evaluation. None of these high- d codes have catalogued transversal pairings with our base-group registry, but the infrastructure now exists to pair them.

VII.2. Limitations

Three limitations frame the numerical claims. First, the $d = 3$ Haar statistics come from either single-sample $t = 10$ draws (Tier-1 panel) or the 100-matrix apples-to-apples panel at $t = 5$ (Section V.3); a joint $t = 10$, ~ 100 -sample Haar run is held up by a qco-kernel limitation that aggregates each subprocess to a single output row regardless of sample size. Second, no $d = 4$ survey is attempted: largest-irrep dimensions scale as t^6 at $d = 4$ and squared matrix-storage as t^{12} , so only the smallest $SU(4)$ subgroups ($|\mathcal{C}| \lesssim 300$) fit in a single-node 125 GB budget at $t = 10$. A pilot on those is tractable overnight; the large- $|\mathcal{C}|$ $SU(4)$ subgroups (e.g. S7F) would need a cluster slot. Third, the target-family T -counts in Table IV are BFS estimates except for the Clifford+ $P(\pi/4)$ row; the $6.8\times$ advantage survives any plausible tightening on the canonical side, but absolute T -counts carry the **bfs_estimate** caveat.

VII.3. Open questions

Two quantitative questions are directly posed by our $t = 5$ data and should be sharp enough to settle at $t \geq 10$: (a) does the four-decimal δ -degeneracy across $\Sigma(216, 648, 1080 \times 3)$ observed for structured diagonal and Howard–Vala extensions in Table II persist, break, or re-rank at larger t ? and (b) does the $\Sigma(360 \times 3)$ apples-panel mean cross strictly below the $\Sigma(216 \times 3)$ mean once the ~ 100 -matrix apples panel removes the 10-matrix outlier sensitivity? Question (b) has been *settled in the affirmative* at $t = 5$ in Section V.3 ($5.7\sigma/\sqrt{N}$ separation at $N = 100$); the $t \geq 10$ rerun will establish whether the same ordering holds at converged design order. Three qualitative questions are left open: (c) a super-golden construction at $d = 3$ saturating $Q_{\text{opt}} \approx 2.69$ on $\Sigma(72 \times 3)$, 2.54 on $\Sigma(216 \times 3)$, or 2.50 on $\Sigma(360 \times 3)$; (d) an explicit data code C_{data} that hosts one of our Σ -series base groups

TABLE V. Magic-state distillation (MSD) status for every extension evaluated in this paper. “Algebra” is the cyclotomic or quaternionic ring hosting the non-Clifford factor of the extension. “Status/closest cousin” combines synthesis and distillation coverage: COMPLETE means transversal code, published MSD protocol, and certified normal-form synthesis are all available; PARTIAL means a transversal code is available but no matching MSD circuit (or the code–extension pairing is not yet established); NONE means no cataloged MSD, either because no protocol exists for the algebra or because the extension is not exactly synthesizable. Rows are grouped by qudit dimension and by the panel section in which the extension appears.

Extension	Algebra	Status / closest cousin	Reference
<i>d = 2 canonical (Section IV)</i>			
Clifford+ $P(\pi/4)$ (canonical $\mathbb{Z}[\omega_8, 1/\sqrt{2}]$)		COMPLETE: BK 15-to-1; Bravyi–Haah triorthogonal	[2, 3]
Clifford+ $P(\pi/8)$	$\mathbb{Z}[\omega_{16}, 1/2]$	NONE: no published protocol for ω_{16} ; cousin BK 15-to-1 [2] does not distill this state	
<i>d = 2 DSA target family (Section VI)</i>			
Clifford+ $P(2\pi/9)$, $P(\pi/9)$, $\mathbb{Z}[\omega_{18}, 1/2]$		PARTIAL (INDIRECT): no direct stabilizer-code distillation (Anderson–Jochym–O’Connor [20]); programmable-MSD distills $ \psi(\theta)\rangle$ on a Clifford+ T substrate; PI code-switching [42] is a non-stabilizer alternative	[38–40, 42]
Clifford+ $P(2\pi/5)$	$\mathbb{Z}[\omega_{10}]$	PARTIAL (INDIRECT): stabilizer-code distillation excluded [20]; programmable-MSD and PI code-switching both apply to arbitrary θ	[38–40, 42]
Clifford+ T_{24} , hurwitz+ T_{12} $\mathbb{Z}[i, \sqrt{2}]$ (super-golden)		PARTIAL: Parzanchevski–Sarnak supplies transversal base; [36] magic state is not a Clifford-orbit stabilizer rotation, so triorthogonal MSD does not apply	
<i>Haar-random rows, any d</i>			
Clifford+rnd, hurwitz+rnd	generic SU(2) (no ring)	NONE: not exactly synthesizable, so MSD is not defined	—
BI+rnd ($d = 2$)	generic SU(2) (no ring)	PARTIAL: Kubischta ((7, 2, 3)) supplies transversal $2I$ base; [17] no MSD for a Haar magic state	
Σ -series+rnd ($d = 3$)	generic SU(3) (no ring)	NONE: not exactly synthesizable; no cataloged qutrit transversal classification for Σ -series	—
<i>d = 3 structured (Section V)</i>			
$\Sigma + \text{HV}(1, 1, 0)$, $\mathbb{Z}[\omega_9]$		PARTIAL: qutrit triorthogonal $[[9m-k, k, 2]]_3$ distills Howard–Vala magic state; Σ -series transversal pairing unestablished	[14]
$\Sigma + \text{HV}(1, 2, 0)$		PARTIAL: cousin Campbell–Anwar–Browne qutrit triorthogonal; no protocol certified for this orbit	[14]
$\Sigma + \text{HV}(2, 1, 0)$	$\mathbb{Z}[\omega_9]$	PARTIAL: qutrit-native ω_9 phase; Campbell–Anwar–Browne applies modulo transversal pairing	[14]
$\Sigma + P(2\pi/9)$ ($d = 3$)	$\mathbb{Z}[\omega_9]$	NONE: ω_{18} outside Campbell–Anwar–Browne; no cousin at $d = 3$	—
$\Sigma + P(\pi/9)$ ($d = 3$)	$\mathbb{Z}[\omega_{18}]$	NONE: ω_{10} not in any qutrit stabilizer setting; no cousin	—
$\Sigma + P(2\pi/5)$ ($d = 3$)	$\mathbb{Z}[\omega_{10}]$		

transversally, analogous to the Denys–Leverrier constructions for $2T$ and $2O$ at $d = 2$ [18, 19] or to the Kubischta–Teixeira twisted unitary t -group framework [41]; and (e) a Ross–Selinger-style normal form for the cyclotomic ring $\mathbb{Z}[\omega_{18}, \frac{1}{2}]$ that would certify the Table IV T -counts.

VII.4. Software release

SWIFTBOT is released with this paper [34] under the MIT license. The release contains: the five-stage pipeline (Section III); the pre-seeded group registry for $d \in \{2, 3, 4\}$; the SWIFTBOT optimizations of the QCO kernel of [32], including the conjugation cache and on-disk $\pi_\lambda(\mathcal{C})$ cache described in Section III; the Stage-4 QEC-code catalog and Stage-5 distillation-protocol catalog with their cross-references to transversal groups; the BFS target-family coverage module used in Section VI; the **q-panel** CLI entry point and the two panel speci-

fications **d3_survey** and **d3_rnd_apples** that reproduce Tables II and III bit-identically; the SQLite cache containing all 54 per-sample rows quoted in Section V; and the L^AT_EX source of this paper. A minimal README walks new users through reproducing Tables I to III on a laptop in under an hour. Pull requests adding new base groups, extensions, codes, or distillation protocols are welcome; the catalog cross-validation tests ensure that every registered transversal-group string resolves to a registered base group.

VIII. CONCLUSIONS

SWIFTBOT—an agentic workflow that ranks $(\mathcal{C}, g)$ pairs by Q_T and cross-references each against a curated fault-tolerance catalog—produces the first systematic $d = 3$ Q_T ranking in our registry and identifies $\Sigma(360 \times 3)$ with a Haar-random extension as the best general-purpose

qudit base ($Q_T/Q_{\text{opt}} \approx 1.32$ in the apples-100 panel at $t = 5$, and consistent single-sample draws at $t = 10$). A paper-scale 10^4 -sample Haar pilot on the binary icosahedral base at $d = 2$ gives $Q_T/Q_{\text{opt}} \approx 1.27$, the natural super-golden candidate. Applied to a lattice-gauge simulation target family, the non-canonical Clifford+ $P(2\pi/9)$ beats canonical Clifford+ T by $6.8\times$ on mean depth despite being worse on Q_T , establishing that the two rankings locate different gates. Only canonical Clifford+ T admits an end-to-end certified fault-tolerant stack; every other winner is covered partially, with sharp open algebraic-compilation or code-pairing problems standing between it and full fault-tolerance. A $d = 4$ pilot on the Fairbairn subgroups [26], an explicit Σ -series $\leftrightarrow$ data-code transversal pairing [19, 41, 42], and a Ross–Selinger normal form for $\mathbb{Z}[\omega_{18}, \frac{1}{2}]$ are the immediate next steps.

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- [1] B. Eastin and E. Knill, Restrictions on Transversal Encoded Quantum Gate Sets, *Phys. Rev. Lett.* **102**, 110502 (2009), [arXiv:0811.4262 \[quant-ph\]](#).
 - [2] S. Bravyi and A. Kitaev, Universal quantum computation with ideal Clifford gates and noisy ancillas, *Phys. Rev. A* **71**, 022316 (2005), [arXiv:quant-ph/0403025](#).
 - [3] S. Bravyi and J. Haah, Magic-state distillation with low overhead, *Phys. Rev. A* **86**, 052329 (2012), [arXiv:1209.2426](#).
 - [4] Y. Wang, Z. Hu, B. C. Sanders, and S. Kais, Qudits and High-Dimensional Quantum Computing, *Front. Phys.* **8**, 589504 (2020).
 - [5] E. O. Kiktenko, A. S. Nikolaeva, and A. K. Fedorov, Colloquium: Quantum Computing with Qudits, *Rev. Mod. Phys.* **97**, 021003 (2025), [arXiv:2212.12934 \[quant-ph\]](#).
 - [6] T. Kim, T. Roy, X. You, A. C. Y. Li, H. Lamm, O. Pronitchev, *et al.*, Ultracoherent superconducting cavity-based multiqutrit platform with error-resilient control, (2025), [arXiv:2506.03286 \[quant-ph\]](#).
 - [7] D. M. Kırkcıoğlu, H. Lamm, O. Ogunkoya, and L. Pierattelli, The $2T$ -quocit: a two-mode bosonic qudit for high energy physics, *arXiv preprint* (2025), [arXiv:2509.21941 \[quant-ph\]](#).
 - [10] M. C. Bañuls *et al.*, Simulating Lattice Gauge Theories within Quantum Technologies, *Eur. Phys. J. D* **74**, 165 (2020), [arXiv:1911.00003 \[quant-ph\]](#).
 - [] Z. Davoudi, I. Raychowdhury, and A. Shaw, Search for efficient formulations for Hamiltonian simulation of non-Abelian lattice gauge theories, *Phys. Rev. D* **104**, 074505 (2021), [arXiv:2009.11802 \[hep-lat\]](#).
 - [] A. Ciavarella, N. Klco, and M. J. Savage, A Trailhead for Quantum Simulation of SU(3) Yang-Mills Lattice Gauge Theory in the Local Multiplet Basis, *Phys. Rev. D* **103**, 094501 (2021), [arXiv:2101.10227 \[quant-ph\]](#).
 - [] J. F. Haase, L. Dellantonio, A. Celi, D. Paulson, A. Kan, K. Jansen, and C. A. Muschik, A resource efficient approach for quantum and classical simulations of gauge theories in particle physics, *Quantum* **5**, 393 (2021), [arXiv:2006.14160 \[quant-ph\]](#).
 - [] P. P. Popov, M. Meth, M. Lewenstein, P. Hauke, M. Ringbauer, E. Zohar, and V. Kasper, Variational quantum simulation of U(1) lattice gauge theories with qudit systems, (2023), [arXiv:2307.15173 \[quant-ph\]](#).
 - [] M. Meth, J. F. Haase, J. Zhang, C. Edmuns, L. Postler, A. Steiner, A. J. Jena, L. Dellantonio, R. Blatt, P. Zoller, T. Monz, P. Schindler, C. Muschik, and M. Ringbauer, Simulating 2D lattice gauge theories on a qudit quantum computer, *Nature Phys.* **21**, 570 (2025), [arXiv:2310.12110 \[quant-ph\]](#).
 - [11] H. Lamm, S. Lawrence, and Y. Yamauchi (NuQS), General methods for digital quantum simulation of gauge theories, *Phys. Rev. D* **100**, 034518 (2019), [arXiv:1903.08807 \[hep-lat\]](#).
 - [12] E. J. Gustafson, H. Lamm, F. Lovelace, and D. Musk, Primitive Quantum Gates for an SU(2) Discrete Subgroup: Binary Tetrahedral, *Phys. Rev. D* **106**, 114501 (2022), [arXiv:2208.12309 \[hep-lat\]](#).
 - [13] B. Assi and H. Lamm, Digitization and subduction of SU(N) gauge theories, (2024), [arXiv:2405.12204 \[hep-lat\]](#).
 - [] J. C. Halimeh, H. Lang, J. Mildenerberger, Z. Jiang, and P. Hauke, Gauge-Symmetry Protection Using Single-Body Terms, *PRX Quantum* **2**, 040311 (2021), [arXiv:2007.00668 \[quant-ph\]](#).
 - [] A. Rajput, A. Roggero, and N. Wiebe, Quantum Error Correction with Gauge Symmetries, *npj Quantum Inf.* **9**, 41 (2023), [arXiv:2112.05186 \[quant-ph\]](#).
 - [] L. Spagnoli, A. Roggero, and N. Wiebe, Fault-tolerant simulation of Lattice Gauge Theories with gauge covariant codes, *Quantum* **10**, 1968 (2026), [arXiv:2405.19293 \[quant-ph\]](#).
 - [14] E. T. Campbell, H. Anwar, and D. E. Browne, Magic-state distillation in all prime dimensions using quantum Reed-Muller codes, *Phys. Rev. X* **2**, 041021 (2012), [arXiv:1205.3104](#).
 - [15] S. Prakash and T. Saha, Low-overhead qutrit magic state distillation, *Quantum* **9**, 1768 (2025), [arXiv:2403.06228 \[quant-ph\]](#).
 - [16] D. Gottesman, Fault-tolerant quantum computation with higher-dimensional systems, *Chaos, Solitons & Fractals* **10**, 1749 (1999), [arXiv:quant-ph/9802007](#).
 - [17] E. Kubischta and I. Teixeira, Family of Quantum Codes with Exotic Transversal Gates, *Phys. Rev. Lett.* **131**, 240601 (2023), [arXiv:2305.07023 \[quant-ph\]](#).

- [18] A. Denys and A. Leverrier, The $2T$ -qutrit, a two-mode bosonic qutrit, *Quantum* **7**, 1032 (2023), [arXiv:2210.16188 \[quant-ph\]](#).
- [19] A. Denys and A. Leverrier, Quantum error-correcting codes with a covariant encoding, *Phys. Rev. Lett.* **133**, 240603 (2024), [arXiv:2306.11621 \[quant-ph\]](#).
- [20] J. T. Anderson and T. Jochym-O'Connor, Classification of transversal gates in qubit stabilizer codes, *Quant. Inf. Comp.* **16**, 0771 (2016), [arXiv:1409.8320 \[quant-ph\]](#).
- [21] A. Y. Kitaev, Quantum computations: algorithms and error correction, *Russ. Math. Surv.* **52**, 1191 (1997).
- [22] N. J. Ross and P. Selinger, Optimal ancilla-free Clifford+ T approximation of z -rotations, *Quant. Inf. Comp.* **16**, 901 (2016), [arXiv:1403.2975 \[quant-ph\]](#).
- [23] V. Kliuchnikov, D. Maslov, and M. Mosca, Fast and efficient exact synthesis of single qubit unitaries generated by Clifford and T gates, *Quant. Inf. Comp.* **13**, 607 (2013), [arXiv:1206.5236 \[quant-ph\]](#).
- [24] M. Howard and J. Vala, Qudit versions of the qubit $\pi/8$ gate, *Phys. Rev. A* **86**, 022316 (2012), [arXiv:1206.1598 \[quant-ph\]](#).
- [25] H. F. Blichfeldt, *Finite Collineation Groups* (University of Chicago Press, 1917).
- [26] W. M. Fairbairn, T. Fulton, and W. H. Klink, Finite and Disconnected Subgroups of $SU(3)$ and their Application to the Elementary-Particle Spectrum, *J. Math. Phys.* **5**, 1038 (1964).
- [27] H. Flyvbjerg, Character table for the 1080 element point group like subgroup of $SU(3)$, *J. Math. Phys.* **26**, 2985 (1985).
- [28] A. Hanany and Y.-H. He, A Monograph on the Classification of the Discrete Subgroups of $SU(4)$, *JHEP* **02**, 027, [arXiv:hep-th/9905212](#).
- [29] W. Grimus and P. O. Ludl, Principal series of finite subgroups of $SU(3)$, *J. Phys. A* **43**, 445209 (2010), [arXiv:1006.0098 \[math-ph\]](#).
- [30] A. Bocharov, M. Roetteler, and K. M. Svore, Factoring with qutrits: Shor's algorithm on ternary and metaplectic quantum architectures, *Phys. Rev. A* **96**, 012306 (2017), [arXiv:1605.02756 \[quant-ph\]](#).
- [31] E. J. Gustafson, H. Lamm, D. Liu, E. M. Muraire, and S. Zhu, Synthesis of Single Qutrit Circuits from Clifford+ R , (2025), [arXiv:2503.20203 \[quant-ph\]](#).
- [32] O. Słowik, P. Dulián, and A. Sawicki, Quantum Circuit Overhead: the cost of approximating unitaries with finite gate sets, *arXiv preprint* (2025), [arXiv:2505.00683 \[quant-ph\]](#).
- [33] P. Sarnak, [Letter to Aaronson and Pollington on the Solovay–Kitaev theorem and golden gates](#) (2015).
- [34] G. T. Fleming, H. Lamm, K. Tame-Narvaez, and P. Vander Griend, SWIFTbot: a pipeline for efficient qudit gate-set discovery, <https://github.com/petergerrit/quai> (2026).
- [35] A. Sawicki and K. Karnas, Universality of Single-Qudit Gates, *Ann. Henri Poincaré* **18**, 3515 (2017), [arXiv:1610.00547 \[quant-ph\]](#).
- [36] O. Parzanchevski and P. Sarnak, Super-golden-gates for $PU(2)$, *Adv. Math.* **327**, 869 (2018), [arXiv:1704.02106 \[math.NT\]](#).
- [37] V. V. Albert and P. Faist, The Error Correction Zoo, <https://errorcorrectionzoo.org> (2024).
- [38] G. Duclos-Cianci and D. Poulin, Reducing the quantum-computing overhead with complex gate distillation, *Phys. Rev. A* **91**, 042315 (2015), [arXiv:1403.5280 \[quant-ph\]](#).
- [39] E. T. Campbell and J. O’Gorman, An efficient magic state approach to small angle rotations, *Quantum Sci. Technol.* **1**, 015007 (2016), [arXiv:1603.04230 \[quant-ph\]](#).
- [40] E. T. Campbell and M. Howard, Magic state parity-checker with pre-distilled components, *Quantum* **2**, 56 (2018), [arXiv:1709.02214 \[quant-ph\]](#).
- [41] E. Kubischta and I. Teixeira, Quantum Codes from Twisted Unitary t -groups, *Phys. Rev. Lett.* **133**, 030602 (2024), [arXiv:2402.01638 \[quant-ph\]](#).
- [42] Y. Ouyang, Y. Jing, and G. K. Brennen, Measurement-free code-switching for low overhead quantum computation using permutation invariant codes, *arXiv preprint* (2024), [arXiv:2411.13142 \[quant-ph\]](#).
- [43] A. Krishna and J.-P. Tillich, Towards low overhead magic state distillation, *Phys. Rev. Lett.* **123**, 070507 (2019), [arXiv:1811.08461 \[quant-ph\]](#).
- [44] A. Wills, M.-H. Hsieh, and H. Yamasaki, Constant-overhead magic state distillation, *Nature Physics* **10.1038/s41567-025-03026-0** (2025).
- [45] L. Golowich and V. Guruswami, Asymptotically Good Quantum Codes with Transversal Non-Clifford Gates, in *57th Annual ACM Symposium on Theory of Computing (STOC ’24)* (2024) [arXiv:2408.09254 \[quant-ph\]](#).
- [46] Q. T. Nguyen, Good binary quantum codes with transversal CCZ gate, in *57th Annual ACM Symposium on Theory of Computing (STOC ’25)* (2025) pp. 697–706.
- [47] M. J. Cervia, H. Lamm, D. Liu, E. M. Muraire, and S. Zhu, Magic State Distillation using Asymptotically Good Codes on Qudits, (2025), [arXiv:2512.21874 \[quant-ph\]](#).